MILITARY SPECIFICATION

MOUNT, RESILIENT, FOR THE M5 BAROSWITCH

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of vibration isolator entitled Mount, Resilient used on the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8803861	- Mount, Resilient.
EL-8871265	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The resilient mount assembly shall comply with all requirements specified on the drawing (dwg) 8803861, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Mock-up unit. The resilient mount assembly is designed to be attached to each of the four mounting surfaces using 6-32 by 9/16 inch filister head machine screws with number 6 steel lock washers under the screw heads. Four resilient mount assemblies are required for each mounting system. A mock-up unit having the dimensions of Figure 1 shall be used for testing the resilient mount assemblies (see 4.4.1.1 through 4.4.1.5).

3.5 Environmental testing. The resilient mount assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The resilient mount assembly shall also comply with 3.5.1 before and after each environmental test. Testing shall be as specified in 4.4.1. Tests shall be conducted in the order specified.

3.5.1 Vibration. The resilient mount assembly shall show no evidence of cracked parts or other damage when subjected to the vibration test (see 4.4.1.1 under the following conditions in Table I.

Table I

Frequency range Hertz (Hz)	Double amplitude displacement (inches) or constant acceleration (g)
4.5 - 23	1.0 g
23.0 - 73	0.036 inch
73.0 - 500	10.0 g

3.5.2 Fatigue. The resilient mount assembly shall show no evidence of broken parts or damage when subjected to vibration at resonance as specified in 4.4.1.2.

3.5.3 Acceleration. The resilient mount assembly shall show no evidence of cracking or splitting when tested as specified in 4.4.1.3.

3.5.4 Shock. The resilient mount assembly shall show no evidence of cracking or splitting when tested as specified in 4.4.1.4. After this test the mounts shall comply with the requirement as specified in 3.5.1.

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3.5.5 High temperature. The resilient mount assembly shall meet the requirements of 3.5.1 after subjection to +200 plus or minus (±) 5 degrees Fahrenheit (°F.) as specified in 4.4.1.5. After this test the mounts shall comply with the requirements of 3.5.1.

3.5.6 Low temperature. The resilient mount assembly shall meet the requirements of 3.5.1 after subjection to -65 ± 5°F as specified in 4.4.1.6. After this test the mounts shall comply with the requirements of 3.5.1.

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 12 complete assemblies (see dwg 8803861) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

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Table II

First article inspection and tests

Inspection or test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Vibration (see 3.5.1)		X
3. Fatigue (see 3.5.2)		X
4. Acceleration (see 3.5.3)		X
5. Shock (see 3.5.4)		X
6. High temperature (see 3.5.5)		X
7. Low temperature (see 3.5.6)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Assembly (see dwg 8803861).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Diameter of bracket holes	Gage
	102. Location of bracket holes	Gage
	103. Pitch diameter of thread	Gage
	104. Thread incomplete or damaged	Visual
	105. Missing component	Visual
	106. Set screw loose or missing	Visual/Manual
	107. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Length of bolt incorrect	Gage
	202. Cushion cups distorted or damaged	Visual
	203. Plating incorrect or damaged to the extent of exposing base metal	Visual
	204. Marking missing, incorrect or illegible	Visual
	205. Spring distorted	Visual
	206. Evidence of poor workmanship (see 3.6)	Visual

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4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packing and packaging not in accordance with MIL-STD-1169(MU)	Visual
102.	Marking not in accordance with MIL-STD-1169(MU)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.6)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Vibration	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Vibration. 100 percent inspection of all mount assemblies shall be conducted until 52 consecutively produced mounts shall have met the requirements of 4.4.1.1. After the qualification of the first 52 consecutively produced resilient mount assemblies have been completed, sample mounts shall be randomly selected at the rate of four per hundred produced. The samples selected shall be subjected to the tests of 4.4.1.1. The failure of any mount to meet the requirements of 4.4.1.1 shall require that 100 percent inspection of all 52 consecutively produced mounts shall have shown the required performance. This 100 percent inspection shall begin with the lot from which the defective samples were selected. Following this, sampling of four per hundred may be resumed.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8871265). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Environmental testing (see 3.5).

4.4.1.1 Vibration test. Attach the resilient mount assemblies to the mock-up unit as indicated in 3.4. The mock-up unit with resilient mount assemblies shall then be mounted on a vibration table and shall be vibrated under the following conditions:

Frequency range Hertz (Hz)	Double amplitude displacement (inches)
10-55-10	0.036 inch

Vibrate the assembly in a direction parallel to the X axis over the above frequency range and determine the frequency at which maximum displacement occurs. Vibrate the assembly at this frequency for 5 minutes. Vibrate the assembly over the frequency range as specified in Table I (see 3.5.1). The transmissibility thus obtained shall fall within the envelope of Figure 2. To assure conformance with Figure 2 data shall be taken at the following frequency points:

Required frequency

(1) 10 Hz
(2) 20 Hz
(3) Primary Resonance Hz
(4) 50 Hz
(5) 60 Hz
(6) 75 Hz
(7) 250 Hz
(8) 500 Hz

Repeat the above tests on the Y and Z axes.

4.4.1.2 Fatigue. The mock-up unit with resilient mount assemblies as indicated in 3.4 shall be vibrated for a period of 3 hours. The unit shall be subjected to 1 hour of vibration along each of the three mutually perpendicular axes. The resilient mount assemblies shall have no broken or damaged parts after completion of this test.

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4.4.1.3 Acceleration. The assembly shall be subjected to a constant acceleration of 50 g for at least 60 seconds applied in both directions along each of three mutually perpendicular axes. The resilient mount assemblies shall have no broken or damaged parts after completion of this test.

4.4.1.4 Shock. Attach the resilient mount assemblies to the mock-up unit as indicated in 3.4 and subject the assembly to 18 impact shocks, 12 of 30 g and 6 of 50 g. One 50 g and two 30 g shocks shall be applied in each direction along the three mutually perpencidular axes, X, Y, and Z. Each 30 g shock impulse shall reach its maximum in 5.5 ± 0.5 milliseconds and shall have a minimum duration of 11.0 milliseconds. Each 50 g shock impulse shall reach its maximum in 2.5 ± 0.5 milliseconds and shall have a minimum duration of 5.0 milliseconds. The amplification of acceleration shall not exceed 2.0 and the resultant deflection of the assembly shall not exceed 0.230 inch. Upon the completion of this test the assembly shall be tested as specified in 4.4.1.1.

4.4.1.5 High temperature. The resilient mount assembly shall be placed in a temperature test chamber and the chamber temperature raised to +200 ± 5°F. This chamber temperature shall be held for a minimum of 2 hours. The resilient mount assembly shall then be reduced to ambient room temperature (75 ± 10°F) without the use of any forced means and shall be tested as specified in 4.4.1.1.

4.4.1.6 Low temperature. The resilient mount assembly shall be tested as outlined in 4.4.1.5 except the test temperature shall be -65 ± 5°F.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The resilient mount assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The resilient mount assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The resilient mount assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

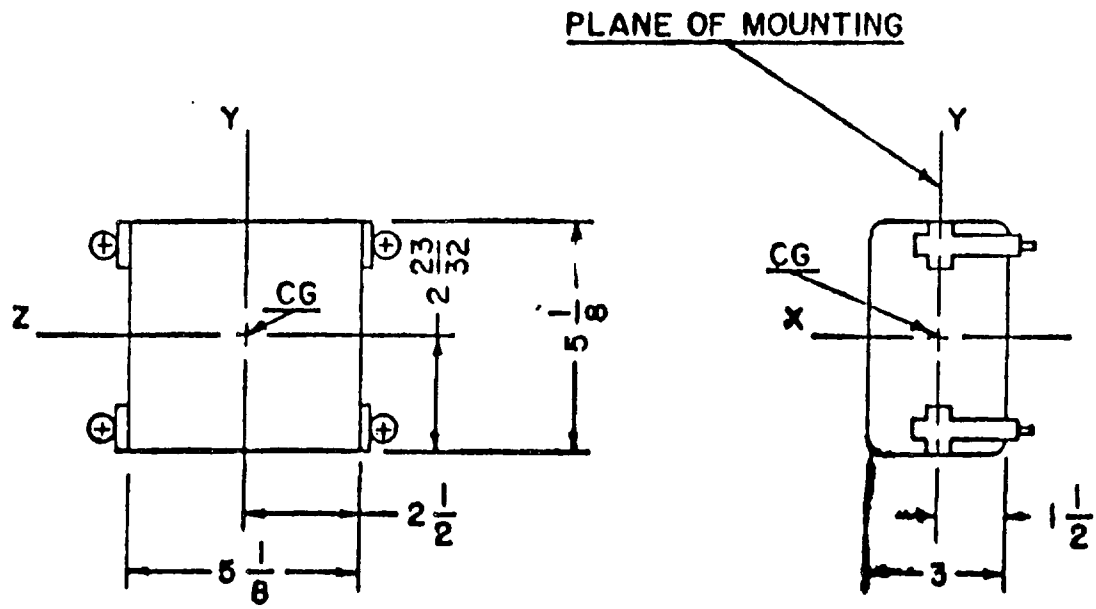
6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1165 (Rev. 1) dated 28 September 1962.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A030

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NOTE:

1. MOCK-UP WGT. = 6 POUNDS, 2 OUNCES ± 1 OUNCE.
2. ALL DIMENSIONS ARE IN INCHES.

Figure 1. Mock-up Unit

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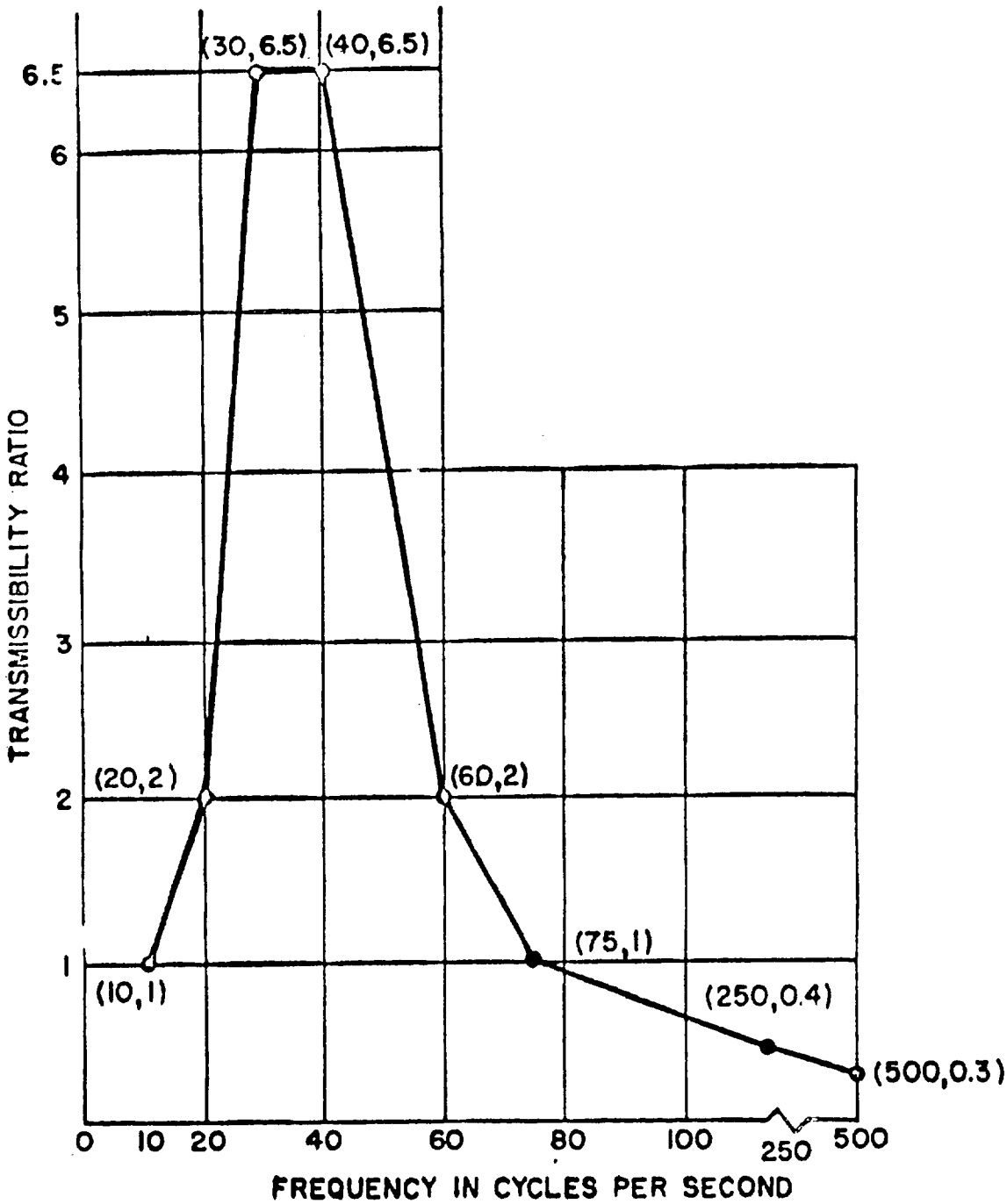


Figure 2. Transmissibility Ratio VS Frequency

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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